

1

a) Mean vector and center

Given $A = (1, 1)$, $B = (2, 2)$, $C = (3, 3)$,calculate the mean vector $\bar{x}$

$$\frac{1}{3} \sum_{i=1}^3 x_i = \left(\frac{1+2+3}{3}, \frac{1+2+3}{3} \right)$$

$$\boxed{\bar{x} = (2, 2)}$$

Centered observations $x_c = x - \bar{x}$; subtracting the mean from each point shifts the data to the origin.

$$A_c = (1-2, 1-2) = (-1, -1)$$

$$B_c = (2-2, 2-2) = (0, 0)$$

$$C_c = (3-2, 3-2) = (1, 1)$$

* It is important to note that centered observations always sum to zero:

$$(-1, -1) + (0, 0) + (1, 1) = (0, 0)$$

b) Centered Data Matrix and Covariance Matrix

Centered data matrix:

$$X_c = \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} A_c \\ B_c \\ C_c \end{bmatrix}$$

Covariance Matrix:

$$S = \frac{1}{n-1} X_c^T X_c = \frac{1}{2} \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\boxed{S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}$$

* Important point to note: All '1' entries indicates a perfect positive correlation between orders and invoices.

c) Eigenvalues and Eigenvectors:

We solve $\det(S - \lambda I) = 0$ for $S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$$(1 - \lambda)^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 2\lambda = 0$$

$$\lambda(\lambda - 2) = 0$$

$$\lambda_1 = 2 \quad \text{and} \quad \lambda_2 = 0$$

Unit Eigenvectors u_j :

For $\lambda_1 = 2$, the Eigenvector satisfies $-u_{11} + u_{12} = 0$

normalizing to length 1, $u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix} \checkmark$

For $\lambda_2 = 0$, $u_{21} + u_{22} = 0$. Normalizing this leads to:

$$u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix} \checkmark$$

* Note: u_1 points along the diagonal where our data lies, capturing 100% of the variance.

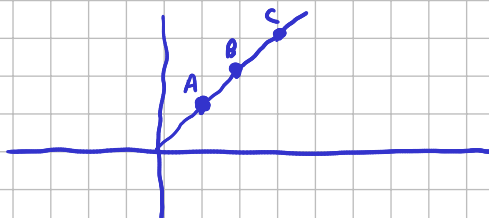
d) First principle component and Eigenvalue analysis

$\Rightarrow$ The first PC is associated with the largest Eigenvalue $\lambda_1 = 2$. The unit vector is:

$$u_1 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

$\Rightarrow$ Explanation as to why the 2nd Eigenvalue is zero:

Geometrically: all observations A, B, C lie on a single line $y = x$ in the 2D plane.



$\Rightarrow$ There is no variance (spread) in the direction perpendicular to this line!

For part (c): $(S - \lambda I)u = 0$ For each λ (Eigenvalue)

$$\left(\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) u = 0$$

$$-u_{11} + u_{12} = 0$$

$$u_{11} = u_{12}$$

End up
with this for
 λ_1

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For λ_2 : $u_{21} = -u_{22} \Rightarrow$ base vector $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

3

$$A = (1, 5), \quad B = (2, 5), \quad C = (4, 5), \quad D = (5, 5).$$

$$\text{Mean vector } \bar{x} = \frac{1}{4} \sum_{i=A}^D x_i = \left(\frac{12}{4}, \frac{20}{4} \right) = (3, 5)$$

Centered observations: $x_c = x - \bar{x}$

$$A: (1-3, 5-5) = (-2, 0)$$

$$B: (2-3, 5-5) = (-1, 0)$$

$$C: (4-3, 5-5) = (1, 0)$$

$$D: (5-3, 5-5) = (2, 0)$$

* Note: All y-components are now zero. This is because the "system-quality score" had zero variance to begin with.

b) Centered data matrix:

$$X_c = \begin{bmatrix} -2 & 0 \\ -1 & 0 \\ 1 & 0 \\ 2 & 0 \end{bmatrix}$$

Covariance Matrix:

$$S = \frac{1}{3} \begin{bmatrix} -2 & 0 \\ -1 & 6 \\ 1 & 0 \\ 2 & 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$$

$$S = \begin{bmatrix} 3.333 & 0 \\ 0 & 0 \end{bmatrix}$$

* Because the second variable (system quality) is constant, its variance and all associated covariances are zero.

$\Rightarrow$ The total variance of the system is contained entirely in the first variable.

c) Eigenvalues and Eigenvectors

characteristic equation

$$\det(S - \lambda I) = 0$$

$$\Rightarrow (3.333 - \lambda)(-\lambda) = 0$$

$$\Rightarrow \lambda_1 = 3.333 \text{ and } \lambda_2 = 0$$

$\Rightarrow$ Eigenvectors: Solve $(S - \lambda I)u = 0$ for all λ

$$(S - \lambda_1 I)u_1 = 0$$

$$\Rightarrow S - \lambda_1 I = \left(\begin{bmatrix} 3.333 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 3.333 & 0 \\ 0 & 3.333 \end{bmatrix} \right) \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & -3.333 \end{bmatrix} \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}$$

$$= 0 u_{11} + 0(u_{12}) = 0$$

$$= 0(u_{11}) - 3.333 u_{12} = 0$$

Not much info!

BUT: Since the matrix is diagonal, the basis vectors (standard) are the eigenvectors.

$$\text{For } \lambda_1: u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \lambda_2: u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

* The first PC u_1 perfectly captures the variance

$$\lambda_1 = 3.333$$

$$\lambda_1 u_1 \checkmark$$

d) Compute the one-dimensional scores on PC1:

$\Rightarrow$ using the formula $z_j = x_c^T u_j$, we project the centered observations onto the first unit Eigvec $u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$z_{1,A} = (-2, 0) (1, 0) = -2$$

$$z_{1,B} = (-1, 0) (1, 0) = -1$$

$$z_{1,C} = (1, 0) (1, 0) = 1$$

$$z_{1,D} = (2, 0) (1, 0) = 2$$

* Note: Since the second variable was constant, the one-dimensional scores on the first PC perfectly preserve the centered values of the digital order activity.

16) A BI team has already centered 3 variables:
revenue growth, service tickets, and self-service logins.

a) Find the Eigenvalues of the covariance matrix:

$$S = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$\Rightarrow$ Solve the characteristic equation $\det(S - \lambda I) = 0$

$$\Rightarrow (4 - \lambda) [(1 - \lambda)^2 - 1] = 0$$

$$(4 - \lambda)(\lambda^2 - 2\lambda) = 0$$

$$\Rightarrow (4 - \lambda)\lambda(\lambda - 2) = 0$$

$$\boxed{\begin{array}{l} \lambda_1 = 4 \\ \lambda_2 = 2 \\ \lambda_3 = 0 \end{array}} \quad \checkmark$$

* The first variable is completely independent of the other two and contains the highest variance ($\lambda_1 = 4$). The other two variables are perfectly correlated, leading to a redundant dimension ($\lambda_3 = 0$).

b) Calculate the Eigenvectors!

$\Rightarrow$ Solve $(S - \lambda I)u = 0$ for each λ .

PC₁ ($\lambda_1 = 4$):

$$\begin{aligned} (S - \lambda I)u_1 &= \begin{bmatrix} 4 - 4 & 0 & 0 \\ 0 & 1 - 4 & 1 \\ 0 & 1 & 1 - 4 \end{bmatrix} \begin{bmatrix} u_{1,1} \\ u_{1,2} \\ u_{1,3} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 1 \\ 0 & 1 & -3 \end{bmatrix} \begin{bmatrix} u_{1,1} \\ u_{1,2} \\ u_{1,3} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$\Rightarrow \text{Row 2: } -3u_{12} + u_{13} = 0$$

$\Rightarrow$ Implication:

$$\text{Row 3: } u_{12} - 3u_{13} = 0$$

$$u_{12} = u_{13} = 0$$

$$\Rightarrow u_{11} \text{ is free } u_{11} = 1$$

$$\Rightarrow u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \checkmark$$

PC₂ ($\lambda_2 = 2$):

$$(S - \lambda I)u_2 = \begin{bmatrix} 4-2 & 0 & 0 \\ 0 & 1-2 & 1 \\ 0 & 1 & 1-2 \end{bmatrix} \begin{bmatrix} u_{21} \\ u_{22} \\ u_{23} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} u_{21} \\ u_{22} \\ u_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Row 1: } 2u_{21} = 0 \Rightarrow u_{21} = 0$$

$$\text{Row 2: } -u_{22} + u_{23} = 0 \Rightarrow u_{22} = u_{23}$$

$$u_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Normalized to unit vector:

$$u_2 = \begin{bmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0.707 \\ 0.707 \end{bmatrix}$$

PC₃ ($\lambda_3 = 0$):

$$(S - \lambda_3 I) u_3 = 0$$

$$\Rightarrow S u_3 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_{31} \\ u_{32} \\ u_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Row 1: } 4u_{31} = 0 \Rightarrow u_{31} = 0$$

$$\text{Row 2: } u_{32} + u_{33} = 0 \Rightarrow u_{32} = -u_{33}$$

$$u_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \quad \text{unit vector: } u_3 = \begin{bmatrix} 0 \\ 0.707 \\ -0.707 \end{bmatrix} \quad \checkmark$$

c) Explained variance ratios: $\lambda_i / \sum \lambda$

$\Rightarrow$ First calculate $\sum \lambda$ (Total variance):

$$\lambda_1 + \lambda_2 + \lambda_3 = 4 + 2 + 0 = \textcircled{6}$$

Ratio for PC₁:

$$\frac{\lambda_1}{6} = \frac{4}{6} = \frac{2}{3} \approx 0.667$$

$$\text{Ratio for PC}_2: \frac{\lambda_2}{6} = \frac{1}{3} \approx 0.333$$

* The first 2 PCs explain 100% of the total variance, confirming that the third dimension is redundant.

d) Scores and Reconstruction Accuracy

$\Rightarrow$ Scores: using $z_j = x_c^T u_j$, for observation $x_c = (2, 1, 1)$

$$z_1 = (2 \ 1 \ 1) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 2 \quad \checkmark$$

$$z_2 = (2 \ 1 \ 1) \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} = \frac{2}{\sqrt{2}} = \sqrt{2} \approx 1.414 \quad \checkmark$$

$\Rightarrow$ Reconstruction: The first two PCs reconstruct x_c exactly!

Justification: the third Eigenvalue λ_3 was found to be zero. The third dimension contains no variance!

$\Rightarrow$ The observation x_c lies within the plane spanned by PC_1 and PC_2 !

16

Nemawashi Trajectories

a)

An online retailer uses a 3-KPI management system over 6 months $t_1, \dots, t_6$. The KPI Series are:

K_1 (Digital Sales share %): 18, 21, 25, 31, 37, 44

K_2 (Avg Basket Value EUR): 52, 53, 52, 56, 60, 65

K_3 (Monthly App Sessions 1000's): 140, 155, 180, 230, 275, 330

$\Rightarrow$ Normalize the 3 KPIs according to the Nemawashi classroom rule, and write the normalized state for each t_i .

$\Rightarrow$ The normalization in the class has 2 phases:

1) Offset adjusted normalization K^*

$$K_t^* = \frac{K_t - K_{\min}}{K_{\max} - K_{\min}}$$

2) Relative share (across time t)

$$K_{j,t}^{**} = \frac{K_{j,t}^*}{\sum_{i=1}^n K_{i,t}^*}$$

$\Rightarrow$ For (1), we adjust with an offset to avoid a zero-sum at t_1 :

$$\text{Range: } \Delta = K_{\max} - K_{\min}$$

$$\text{Offset Baseline: } B = K_{\min} - (0.1 \Delta)$$

$$\text{Adjusted: } K_t^* = \frac{K_t - B}{\Delta}$$

Time	K_1^*	K_2^*	K_3^*
t_1	$\frac{18 - [18 - (0.1 \cdot 26)]}{26}$	$\frac{52 - [52 - (0.1 \cdot 13)]}{13}$	$\frac{140 - [140 - (0.1 \cdot 190)]}{190}$
	= 0.100	= 0.100	= 0.100
t_2	0.215	0.177	0.179

t_3	0.369	0.100	0.311
t_4	0.600	0.408	0.574
t_5	0.831	0.715	0.811
t_6	1.100	1.100	1.100

For (2) we proceed as usual (HH's lecture):

Time	K_1^{**}	K_2^{**}	K_3^{**}
t_1	$\frac{0.100}{0.1 + 0.1 + 0.1}$ $= 0.333$	" "	" "
t_2	0.377	0.310	0.313
t_3	0.473	0.128	0.399
t_4	0.379	0.258	0.363
t_5	0.353	0.303	0.344
t_6	0.333	0.333	0.333

b) Plot the Nemanashi trajectory

=> Homework!

c) Identify one phase in which all 3 KPIs improve together and one phase in which one KPI lags behind the other two.

=> In the phase $t_5 \rightarrow t_6$, all 3 KPIs increase together to their max values, since bigger is better in this scenario, it meets the condition of all 3 improving together.

=> In the phase $t_2 \rightarrow t_3$, K_2 lags behind K_1 and K_3 . In the raw data K_2 is relatively flat, but drops sharply when looking at K_2^{**} !